Obstacles in C-space

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Where this fits:

- **Builds on:** Vector magnitude, distance and scalar operations; Maps and landmarks; Configuration space.
- **Leads to:** Collision detection; Distance queries and GJK; Bug algorithms; Preimage planning; Inevitable collision states.
- **Compare with:** Breadth-first search.

1. Overview

Key point: A placement of the robot is free only if its whole body misses every obstacle, not just its centre. Drawing each obstacle in C-space as the set of placements that touch it (for a round robot, the obstacle grown by the robot's radius) turns "is the robot clear?" into "is this point outside the grown obstacles?".

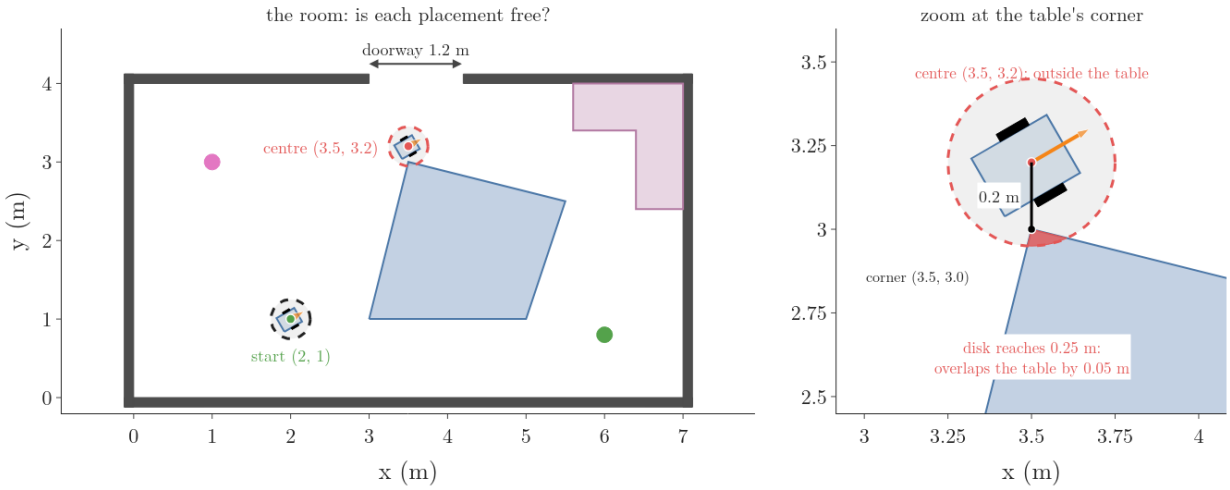


Figure 1: Left: the room with the robot's disk footprint, 0.25 m in radius, at the start (2, 1) and at a second placement whose centre is (3.5, 3.2); the doorway in the top wall is 1.2 m wide. Right: a zoom on the table's corner (3.5, 3.0): the centre is 0.2 m above the corner, outside the table, but the disk reaches 0.25 m and overlaps the table by 0.05 m (red)

So far the robot can plan a route on the room's grid: breadth-first search reaches the cell under the door in 8 moves, and A* finds a 3.414 m route with diagonal moves. The grid routes move a point. The **configuration space** (G-2287), C-space for short, makes the whole robot a point too: each placement (x, y, θ), position and heading, is one point, and the basic motion planning problem is to find a path of free points from the start, (2 m, 1 m) facing 30°, to the doorway centre, (3.6 m, 4.0 m).

The remaining problem is the meaning of the word "free". The map of the room marks where the obstacles are, but the robot is not a point. Its footprint, the floor it covers, is a disk of radius 0.25 m centred on the middle of its axle (Figure 1):

- with its centre at (3.5, 3.2), the robot stands 0.2 m above the table's corner (3.5, 3.0); the centre is outside the table, yet the disk reaches 0.25 m and overlaps the table by 0.05 m;
- the doorway is 1.2 m wide, but a robot 0.5 m wide cannot put its centre close to either side of it;
- if the robot carries a board 1.4 m long, which way it faces decides whether the board fits through the 1.2 m doorway at all.

The opening question of this Note: **which placements (x, y, θ) are free, so that a planner can search C-space for a route to the door?** We answer it in steps:

- the placements that touch an obstacle, drawn in C-space, and the free placements that remain, for the round robot (Section 2);
- a rule that works for any shape, such as the board, and what turning does to it (Section 3);
- whether a path can exist at all: the separate pieces of free space (Section 4).

2. The obstacle region and free space

> **Key point:** The **obstacle region** (G-2650) is every placement at which the robot touches an obstacle; **free space** (G-2651) is all the others. For the disk robot, a placement touches an obstacle exactly when its centre is within 0.25 m of it, so the obstacle region is every obstacle grown by 0.25 m.

2.1 Why the centre alone cannot say what is free

The first part of the opening problem is the placement with its centre at (3.5, 3.2). Its centre is outside the table, so a planner that tests only the centre calls it free. Figure 1 shows that it is not: the disk reaches below the centre to y = 2.95, and the table's corner is at y = 3.0. With:

- the centre's height, 3.2 m;
- the robot's radius, 0.25 m;
- the corner's height, 3.0 m.

The lowest point of the disk, straight below the centre:

$$3.2 - 0.25 = 2.95 \text{ m}$$

The disk's lowest point lies below the corner, by:

$$3.0 - 2.95 = 0.05 \text{ m}$$

So whether a placement is free depends on every floor point the robot covers, not on one point. Testing the whole disk against every obstacle for every placement a planner tries is

the slow way; the rest of this section moves the test to the centre, where a planner can make it in one step.

2.2 The obstacle region: every placement that touches an obstacle

To move the test to the centre, we picture the robot sliding round the table while it keeps touching it, like a coin pushed round the edge of a book. Its centre traces a closed curve round the table (Figure 2). Every centre inside that curve puts the disk on the table; every centre outside leaves a gap. So the curve splits the placements in two, and we can draw the split once, in C-space, instead of testing the disk again and again.

Figure 2 plays the slide, counter-clockwise from under the table's bottom edge. Watch the stages named in its titles. "Along an edge": the centre runs parallel to the edge, 0.25 m out. "At a corner": the centre swings round the corner on an arc of radius 0.25 m. The last stage, "the centre (3.5, 3.2) lies inside the grown table", fills the curve and marks the placement of Figure 1, which falls inside.

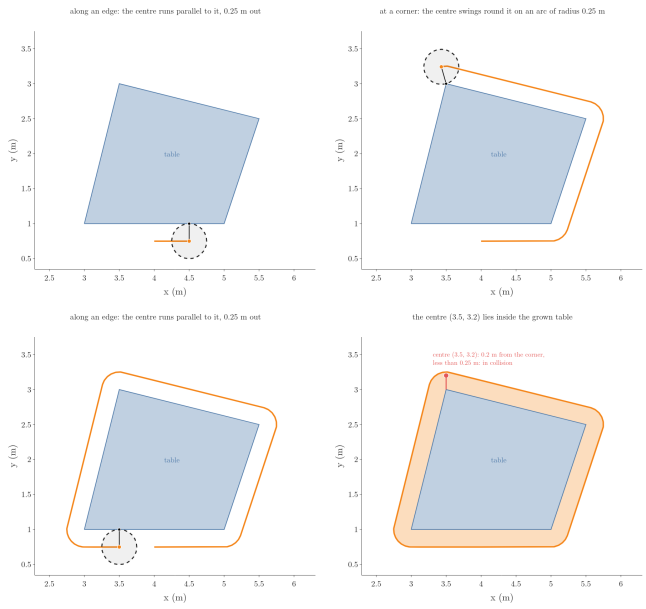


Figure 2: Animation: the robot's 0.25 m disk footprint slides counter-clockwise round the table, always touching it; a black line joins its centre to the touching point. Along an edge the centre runs parallel to the edge, 0.25 m out; at a corner it swings round on an arc of radius 0.25 m. The orange trace of the centre closes into the boundary of the grown table, which the last frames fill (orange) and in which they mark the centre (3.5, 3.2), 0.2 m from the corner, in collision

The standard terms (LaValle §4.3.1; Hauser ch.9):

- The set of floor points the robot covers when it is placed at a configuration (a placement) is its footprint at that **configuration** (G-2286).

- The set of all configurations at which the footprint touches or overlaps an obstacle is the **obstacle region** (G-2650), also called the C-obstacle, or C-space obstacle.
- Everything else in C-space is **free space** (G-2651): the configurations at which the robot touches nothing.

The obstacle region of the table, the filled shape in the last frame of Figure 2, is the table's C-obstacle. The configuration whose centre is (3.5, 3.2) lies in it; the start, (2, 1), does not.

The formal version needs set notation, which we read on small instances:

- q is a configuration, such as the start, $q = (2\text{ m}, 1\text{ m}, 30°)$;
- C is the C-space, the set of all configurations;
- $A(q)$ is the robot's footprint at q, a set of floor points: at the start, every point within 0.25 m of (2, 1);
- O is the obstacles, the set of floor points covered by the table, the cabinet, the poles and the walls;
- $\cap$ gives the intersection (G-967), the points in both sets, and $\emptyset$ is the empty set, so $A(q) \cap O \neq \emptyset$ reads "the robot and the obstacles share at least one point";
- $\setminus$ removes one set from another: $C \setminus C_{\text{obs}}$ is C without the obstacle region;
- $\in$ reads "is in": $q \in C$ says that q is one of the configurations in C;
- in the braces, the bar reads "such that".

$$C_{\text{obs}} = \{q \in C \mid A(q) \cap O \neq \emptyset\}$$

$$C_{\text{free}} = C \setminus C_{\text{obs}}$$

(LaValle §4.3.1, eq. 4.34; Hauser ch.9). A planner now asks one question per configuration: is q in C_{obs}? For the disk, the slide of Figure 2 already gave the answer's shape; the next step gives its rule.

2.3 Growing the obstacles by the robot's radius

> **Key point:** For a disk robot, the obstacle region of each obstacle is a **grown obstacle** (G-2652): the obstacle with its edges pushed out by the radius, 0.25 m, and its corners rounded with arcs of radius 0.25 m. The grown table covers 5.747 m^2 against the table's 3.625 m^2.

Figure 2 found the table's obstacle region by sliding the disk round it. A planner needs a rule it can compute for every obstacle of the room, and the disk gives a simple one.

Why the rule holds. The disk covers every floor point within 0.25 m of its centre. The disk touches an obstacle exactly when some obstacle point lies within 0.25 m of the centre: when the centre's Euclidean distance (G-715), the straight-line distance, to the nearest obstacle point is at most 0.25 m. So for the disk (Hauser ch.9, "Planar translating disks"):

- r is the robot's radius, 0.25 m;

- $\mathrm{dist}(q, O)$ is the distance from the centre of configuration q to the nearest point of the obstacles O, in m: 0.2 m for the centre (3.5, 3.2).

$$C_{\text{obs}} = \{q \mid \mathrm{dist}(q, O) \leq r\}$$

The configuration of Figure 1 passes the test:

$$\text{dist} = 0.2 \leq 0.25$$

so it is in the obstacle region, as the disk's 0.05 m overlap showed. An obstacle grown this way, every point within r of it added, is a **grown obstacle** (G-2652), also called an inflated obstacle. The robot has become a point, and the obstacles have grown by its size.

Why lines and arcs. Along an edge, the nearest obstacle point is on that edge, so the centres at distance 0.25 m form a line parallel to the edge. Round a corner, the nearest point is the corner itself, so they form an arc of a circle of radius 0.25 m about the corner (Hauser ch.9). Each arc spans the angle by which the table's outline turns at that corner. At the corner (3.5, 3.0) the two edges meet at a right angle, so its arc is a quarter circle. A centre straight above that corner, like the one of Figure 1, has the corner as its nearest table point, 0.2 m away.

A check: the grown table's area. The growth adds a strip along every edge and a wedge of a circle at every corner. The table's numbers (notebook):

- the table's area, 3.625 m^2;
- the table's perimeter, the four edges added, 7.704 m;
- the turns of its outline at its four corners: 104.0°, 71.6°, 94.4° and 90.0°.

Walking round a convex outline, we turn at each corner and finish facing the way we started, so the turns add up to one full turn:

$$104.0 + 71.6 + 94.4 + 90.0 = 360^\circ$$

The four corner wedges therefore make one whole circle of radius 0.25 m. The strips are each edge's length times 0.25 m. Adding the parts:

$$\text{grown area} = 3.625 + 7.704 \times 0.25$$

$$+\pi \times 0.25^2$$

$$= 3.625 + 1.926 + 0.196$$

$$= 5.747 \text{ m}^2$$

The same steps in symbols:

- s is the obstacle's area, in m^2;
- p is its perimeter, in m;
- r is the robot's radius, in m.

$$\text{grown area} = s + p\,r + \pi r^2$$

The library that grows shapes in the notebook (`shapely`) measures the same 5.747 m^2. So the table's obstacle region is 2.122 m^2 larger than the table:

$$5.747 - 3.625 = 2.122\ \mathrm{m}^2$$

Every one of those extra square metres is floor that looks free on the map but where the robot's centre cannot go. The table is one obstacle; the room has four more kinds.

2.4 The room in C-space, and the drive to the door

> **Key point:** When every obstacle of the room, walls included, is grown by 0.25 m, what stays white is the free space of the disk robot. The doorway leaves 0.7 m for the centre, and the straight 3.4 m drive from the start to the doorway centre clears the grown table by 0.132 m.

The second part of the opening problem is the doorway. Growing every obstacle answers it and the first part at once. Figure 3 grows the whole room (Hauser ch.9):

- the walls move 0.25 m into the room, because the centre must stay 0.25 m from each wall;
- each pole, 0.1 m in radius, becomes a disk 0.35 m in radius;
- the cabinet's L grows along its outline;
- the door jambs, the wall ends at (3.0, 4.0) and (4.2, 4.0), grow into quarter-disks of radius 0.25 m.

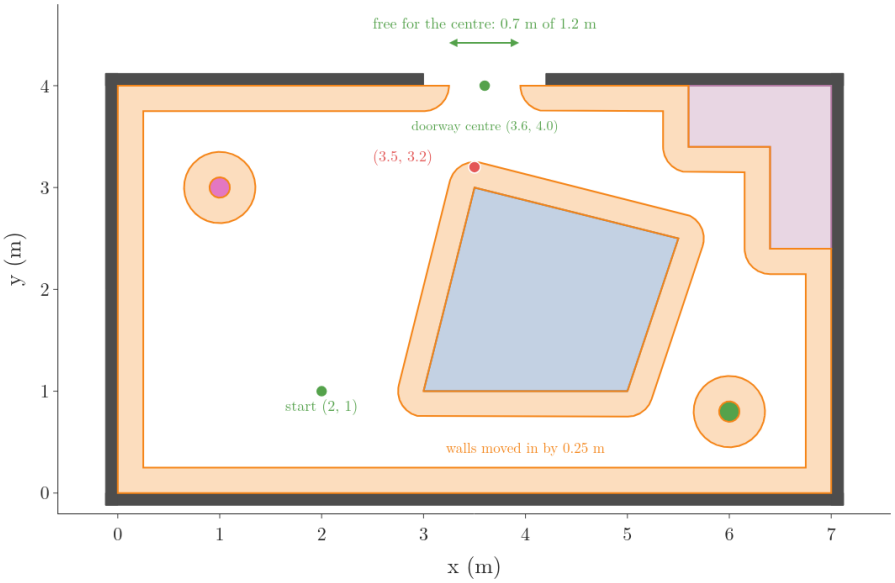


Figure 3: The obstacle region of the room for the disk robot: every obstacle grown by 0.25 m (orange), the walls moved 0.25 m inwards, the poles grown to 0.35 m disks, the door jambs grown into quarter-disks. White is the free space. The centre (3.5, 3.2) (red) is inside the grown table; the start (2, 1) and the doorway centre (3.6, 4.0) are free; only 0.7 m of the 1.2 m doorway is free for the centre

In Figure 3, white is free space and orange is the obstacle region; a configuration is free when its centre is on white. The doorway's free stretch lies between the two grown jambs:

$$3.0 + 0.25 = 3.25 \text{ m}$$

$$4.2 - 0.25 = 3.95 \text{ m}$$

$$3.95 - 3.25 = 0.7 \text{ m}$$

So of the 1.2 m doorway, 0.7 m is open to the centre, which settles the second part of the opening problem.

The drive in both pictures. Now the robot can check a whole route. The simplest is the straight drive from the start (2, 1) to the doorway centre (3.6, 4.0): the robot turns on the spot to face along it and drives forward. The drive runs 1.6 m along x and 3.0 m along y:

$$3.6 - 2.0 = 1.6 \text{ m}$$

$$4.0 - 1.0 = 3.0 \text{ m}$$

$$\text{length} = \sqrt{1.6^2 + 3.0^2} = 3.4 \text{ m}$$

$$\text{heading} = \text{atan2}(3.0,\ 1.6) = 61.9^\circ$$

where atan2 (G-2325) turns the rise, 3.0 m, and the run, 1.6 m, into the direction of the drive.

Figure 4 plays the drive in two pictures side by side, the way it is usually drawn (MR 10.2). Left, the room: the robot turns, then drives. Right, C-space seen from above: the same drive is a point moving through the white free space. Watch the frame "closest to the table": the robot's centre passes 0.382 m from the table's corner, and the point passes 0.132 m from the grown table. The two numbers are one fact in two pictures:

$$0.382 - 0.25 = 0.132 \text{ m}$$

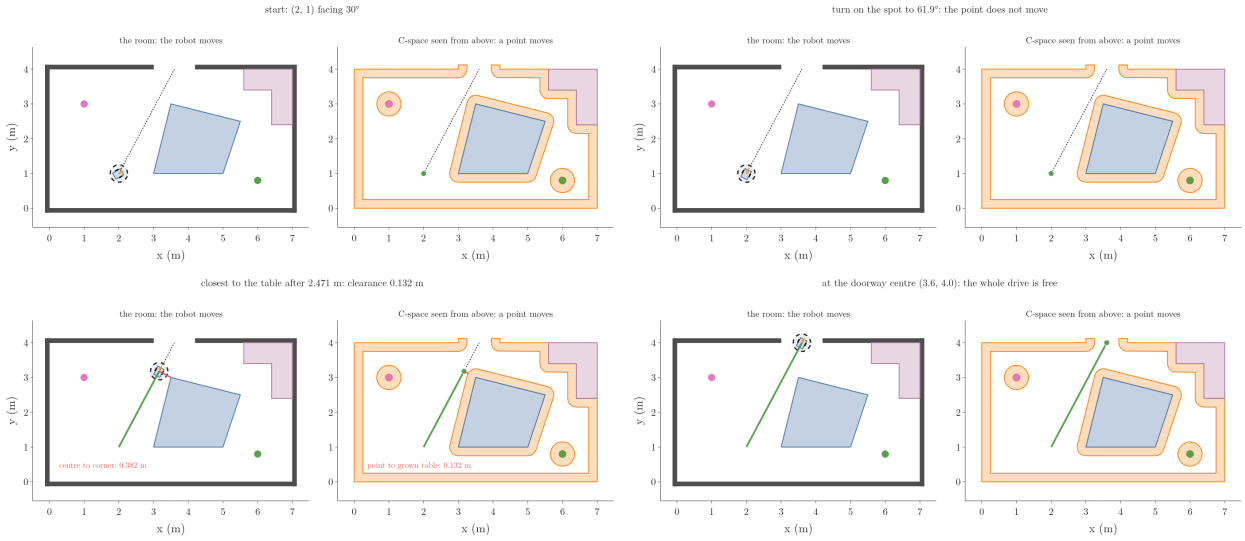


Figure 4: Animation in two panels. Left, the room: the robot (disk footprint, radius 0.25 m) at the start (2, 1) facing 30 degrees turns on the spot to 61.9 degrees, then drives straight 3.4 m to the doorway centre (3.6, 4.0); its path so far is green. Right, C-space seen from above: the obstacles grown by 0.25 m (orange) and the robot as a point moving along the same line; during the turn the point does not move. Frame "closest to the table": after 2.471 m the centre is 0.382 m from the table's corner, and the point is 0.132 m from the grown table. The last frame shows the whole drive inside free space

The disk is round, so turning on the spot changes nothing it covers: in Figure 4 the point stands still while the robot turns. Every heading gives the same picture, so the C-space slice drawn on the right describes the whole C-space for the disk. The straight drive stays in free space all the way, so the disk robot can drive straight to the door with 0.132 m to spare. The

rule that gave this answer, "grow by the radius", used one fact about the disk: it looks the same from every side. The board of the opening problem does not.

3. The Minkowski sum: obstacles for any robot shape

> **Key point:** For a robot of any shape that moves without turning, the obstacle region (the placements that touch an obstacle) is the **Minkowski sum** (G-2653) of the obstacle and the robot's shape reflected through its centre: every obstacle point minus every robot point. For two convex polygons it is a convex polygon with at most as many corners as both together.

3.1 Why a board breaks the disk rule

The third part of the opening problem: the robot carries a board 1.4 m long and 0.3 m wide, centred over the middle of its axle and lying along its heading. The board is what now touches the obstacles, so its outline is the footprint; the 0.25 m disk lies inside it. The board is a new example for one reason: it is not round, so the direction the robot faces matters.

Facing along the wall, at heading 0°, the board lies across the doorway with its full 1.4 m:

$$1.2 - 1.4 = -0.2 \text{ m}$$

The board is 0.2 m too long to fit. Facing the door, at 90°, only its 0.3 m width crosses:

$$1.2 - 0.3 = 0.9 \text{ m}$$

So "grow every obstacle by one radius" cannot work: the board reaches 0.7 m ahead of its centre but only 0.15 m to the side. We need a rule that builds the obstacle region from any shape, then we can turn the shape.

3.2 Minkowski sum and difference

The rule starts from one placement on the edge of the board's obstacle region.

Where the board touches. Fix the heading at 0° and slide the board towards the table until its front end touches the table's bottom-left corner (the first frame of Figure 5). The configuration gives the position of one chosen point of the robot, the middle of its axle, which is also the board's centre. Each point of the board sits at a fixed offset from that centre: the middle of the front end is 0.7 m ahead. The touching point is that offset added to the centre, and it equals the corner. With the numbers, the centre q is:

$$q + (0.7,\ 0) = (3.0,\ 1.0)$$

$$q = (3.0,\ 1.0) - (0.7,\ 0)$$

$$q = (2.3,\ 1.0)$$

The same steps in symbols, with:

- q is the position of the centre, (2.3, 1.0) here;
- a is the offset of a point of the robot from the centre, (0.7, 0) for the middle of the front end;
- o is the obstacle point it touches, the corner (3.0, 1.0);
- A is the robot's shape as a set of offsets, the board at heading 0°;
- O is the obstacle as a set of points, the table.

$$q + a = o$$

$$q = o - a$$

Touching means some robot point lands on some obstacle point, so the centre is the obstacle point minus the robot point's offset. Collecting every such centre over every obstacle point and every robot offset gives the obstacle region of the robot moving without turning:

$$C_{\text{obs}} = \{o - a \mid o \in O,\ a \in A\}$$

$$= O \ominus A$$

(LaValle §4.3.2, eq. 4.37; Hauser ch.9; KAIST C-space slides). Taking every difference of points of two sets is the **Minkowski difference** (G-2654), written $\ominus$. Taking every sum instead is the **Minkowski sum** (G-2653), written $\oplus$, "add every point of one shape to every point of the other". Subtracting a is adding $-a$, the offset turned through 180° about the centre. So with $-A$, the robot reflected through its centre:

$$O \ominus A = O \oplus (-A)$$

(LaValle §4.3.2). The board and the disk are both symmetric about their centre: each offset a has a partner $-a$ on the far side (the front end's middle (0.7, 0) and the back end's middle (−0.7, 0)). So for them the reflected shape is the shape itself, and the obstacle region is the obstacle plus the robot. For the disk, adding a 0.25 m disk to every table point gives every point within 0.25 m of the table: the grown table of Section 2.3 is a Minkowski sum.

Building it. "Add every point of one shape to every point of the other" is a picture of sweeping one shape over the other (Hauser ch.9, "Minkowski sums and differences"): place a copy of the reflected board, centred, on every point of the table, and the copies cover the obstacle region. Figure 5 builds it in that order. Watch the stages named in its titles: the board touching the corner; a reflected board centred on each table corner in turn; copies

along the edges; copies everywhere, which sweep out a region; and, in the last frame, its outline.

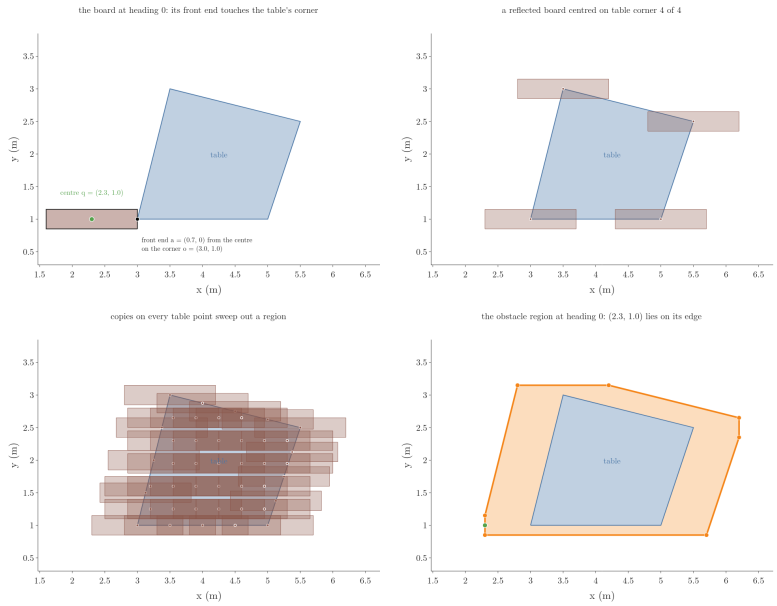


Figure 5: Animation: the board's obstacle region at heading 0 as a Minkowski sum. First the board (1.4 m by 0.3 m, brown) with its front end on the table's corner (3.0, 1.0): its centre q is at (2.3, 1.0). Then copies of the reflected board are centred on the table's four corners, then along its edges, then inside it; together they sweep out a region. The last frames outline it (orange): a convex polygon with 7 corners, on whose left edge the centre (2.3, 1.0) lies

The last frame of Figure 5 has 7 corners. Every corner of the outline is a table corner plus a board corner, and every edge of the outline is a table edge or a board edge moved into place (LaValle §4.3.2, "A polygonal C-space obstacle"; KAIST C-space slides). So for two convex (G-479) polygons, sets that contain the segment between any two of their points, the Minkowski sum is convex and has at most as many corners as both together (Hauser ch.9; KAIST C-space slides):

$$4 + 4 = 8 \text{ corners at most}$$

At heading 0° the table's bottom edge and the board's long edges are all horizontal, so two edges line up into one and 7 corners remain; at 30° there are 8 (notebook). A recipe a computer can follow (Hauser ch.9): add each of the 4 board corners (reflected) to each of the 4 table corners, 16 points, and take their convex hull (G-477), the smallest region with straight edges that contains them. The cabinet, a union of two rectangles, gets one Minkowski sum per rectangle, joined (LaValle §4.3.2). So one rule now covers any shape that keeps its heading; the board, though, turns with the robot.

3.3 Turning: a different slice for every heading

> **Key point:** Each heading gives its own obstacle region in the plane, a slice of C-space; stacked by heading they make the obstacle region in (x, y, θ). For the board the slices change shape: the doorway is closed to the centre at 0° and open over 0.90 m at 90°.

The board's obstacle region of Figure 5 was for heading 0° only. When the robot turns to 30°, the board points a different way, so the reflected board in the Minkowski sum turns too, and the obstacle region changes shape. The obstacle region for all configurations (x, y, θ) is three-dimensional; holding the heading fixed cuts it into a flat slice: the set of positions (x, y) that are free, or not, at that one heading (KAIST C-space slides; Hauser ch.9, "C-obstacles for rotational motion").

Figure 6 plays the slices of the whole room for the robot with the board, one heading per frame, from 0° to 180° in steps of 10°. The small panel shows which way the board points. Watch the orange obstacle region stretch along the board's direction as it turns, and watch the green bar in the doorway: the doorway's free stretch for the centre.

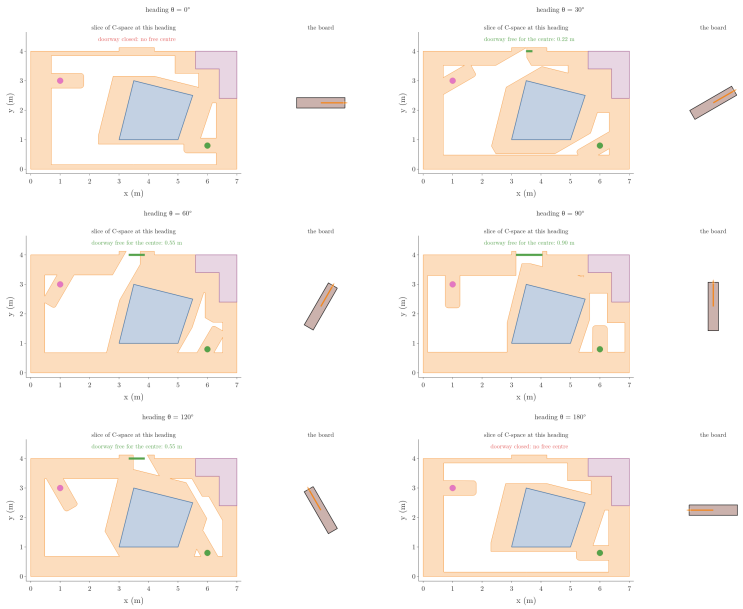


Figure 6: Animation: the obstacle region of the room (orange) for the robot carrying the 1.4 m by 0.3 m board, one heading per frame from 0 to 180 degrees in steps of 10 degrees; white is free; a small panel shows the board's heading. The orange region stretches along the board's direction. The doorway's free stretch for the centre (green bar): closed at 0 degrees, 0.22 m at 30, 0.55 m at 60, 0.90 m at 90, then closing again; the slice at 180 degrees is the same as at 0

The doorway's free stretch at each heading (notebook):

Heading	0°	20°	30°	60°	90°
Free stretch of the doorway	0	0.05 m	0.22 m	0.55 m	0.90 m

At 0° the doorway is shut, as the 0.2 m shortfall of Section 3.1 said; it opens at about 18° and is widest, 0.90 m, when the board faces the door. At 180° the board covers the same floor as at 0°, so the slice is the same: the board's obstacle region repeats every half turn.

Figure 7 stacks the slices with the heading as the height, for the table alone. For the board, the slices turn and change shape as the heading rises, so the obstacle region is a twisted column. For the disk, every slice is the same grown table, so the column is straight. Both columns run from 0° to 360°, and in C-space the top and the bottom are the same heading (360° is 0°), so each column's top joins its bottom: one region, not two ends (MR 10.2).

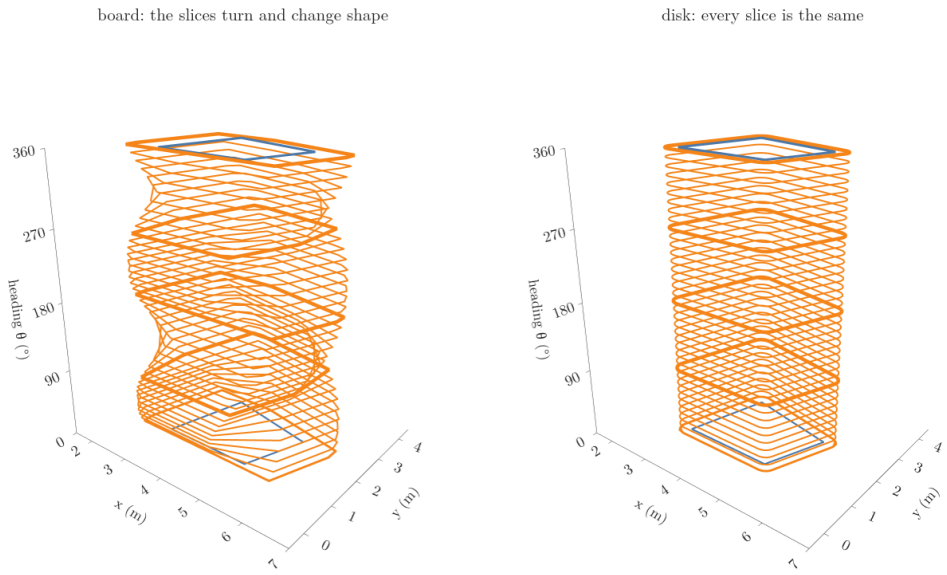


Figure 7: Slices stacked by heading. Left: the table's obstacle region for the board, one orange outline every 10 degrees of heading from 0 to 360 (thicker every 90 degrees); the outlines turn and change shape, making a twisted column. Right: the same for the disk robot; every outline is the table grown by 0.25 m, making a straight column. The table itself is drawn in blue at the bottom and the top

So for the board, whether a position is free depends on the heading: the centre (3.6, 4.0) in the doorway is in the obstacle region at 0° and free at 90°. The robot carrying the board must face the door to pass. Free space has become a three-dimensional shape that may be cut into separate pieces, which decides whether any route exists.

4. Connected components of free space

> **Key point:** Free space, the placements that touch nothing, can fall into separate pieces, its **connected components** (G-2656), and a path between two configurations exists only if both lie in the same piece. For the disk the room's free space is one piece; for the board single slices split into 2 to 4 pieces, but turning joins them all into one.

4.1 One piece or several

The opening question asks for free placements so that a planner can search for a route. Before any search, one question settles whether a route can exist: can the robot get from the start to the door without leaving free space?

Free space can fall into pieces that free paths cannot join. Within one piece, the robot can move from any configuration to any other without touching an obstacle; between two pieces, every path crosses the obstacle region. Each such piece is a **connected component** (G-2656) of free space, and a path between two configurations exists only if both lie in the same component (MR 10.2).

The disk robot. In Figure 3 the white free space is one piece: every gap between grown obstacles stays open. The narrowest is between the green pole and the bottom wall:

- the pole's centre is 0.8 m above the wall;
- the pole's radius is 0.1 m;
- the robot is 0.5 m wide.

The gap under the pole, and what is left of it once the robot is in it:

$$0.8 - 0.1 = 0.7 \text{ m}$$

$$0.7 - 0.5 = 0.2 \text{ m}$$

The notebook counts the pieces of the free space on a grid of 0.01 m and finds one. So the start and the door are in the same component, which is why the drive of Figure 4 could exist.

4.2 Turning joins the pieces

The board, one heading at a time. The disk's free space is one piece; the board's is not, at a fixed heading. Figure 8 colours the free pieces of two slices, one colour per piece. At heading 0° the free space has 2 pieces: a large one and a small one of 0.39 m^2 right of the table, above the green pole. At heading 90° there are 3:

- a large piece that holds both the start (2, 1) and the doorway centre (3.6, 4.0);
- a piece round the green pole, which holds the point (6.6, 1.3);
- a sliver of 0.004 m^2 between the table and the cabinet.

At 60° there are 4 (notebook).

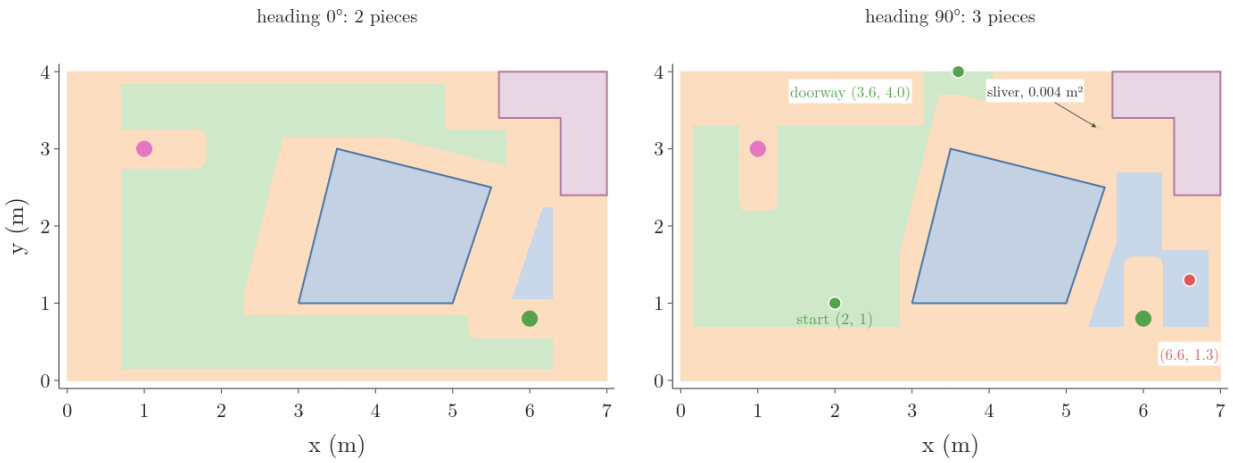


Figure 8: Connected components of free space in two slices for the robot carrying the board; each colour is one piece of free centres, orange is the obstacle region. Heading 0 degrees: 2 pieces. Heading 90 degrees: 3 pieces; the start (2, 1) and the doorway centre (3.6, 4.0) share the large green piece, the point (6.6, 1.3) is in the blue piece round the green pole, and a sliver of 0.004 square metres sits between the table and the cabinet

A robot at (6.6, 1.3) facing the door, if it kept that heading, could never reach the door: its piece of the 90° slice does not include the doorway. The robot can turn, though, and a turn moves it up or down through the stack of Figure 7 into another slice, where the pieces lie differently. The notebook joins all the slices from 0° to 179°, every 1°, with the heading wrapping round, and counts the pieces of free space in (x, y, θ): **one**. The pieces of each slice are cross-sections of that one region, cut at one heading, as an obstacle drawn on the 2R arm's square C-space can look like three pieces until we remember that its opposite edges are glued (MR 10.2).

So for the board too a path exists from the start to the door. The robot can turn on the spot at the start from 30° to 90°, which stays free at every heading in between (notebook), and the large piece of the 90° slice then holds both ends. A path in C-space may also move sideways, which a differential-drive robot cannot do directly (no sideways motion); turning such a path into wheel commands is a later topic.

4.3 Why planners do not build the obstacle region

The obstacle regions of this Note were cheap to build: a disk and a board against a few convex shapes in a 3D C-space. Real robots break all three conditions:

- the robot and the obstacles are not convex; a Minkowski sum of two non-convex shapes is the union of the sums of their convex pieces and can have many more corners (Hauser ch.9);

- in 3D, two objects of 444 and 1,134 triangles give a Minkowski sum made of 66,667 pieces (KAIST C-space slides 15 and 16);
- an arm with six joints has a six-dimensional C-space, in which no picture or stack of slices can hold the obstacle region.

So building the obstacle region exactly is impractical (MR 10.2; KAIST C-space slides). Most planners never build it. Instead, a planner asks a subroutine one question at a time: is this configuration free, and is this short path segment free (MR 10.2; Hauser ch.10, "Implicit C-obstacle representation")? Answering those questions fast, for thousands of configurations, is collision checking.

5. Summary

Idea	What it is	On our room
Obstacle region	configurations at which the robot touches an obstacle	centre (3.5, 3.2) is in it, 0.2 m from the table
Free space	all other configurations	white in Figure 3
Grown obstacle	for a disk robot, the obstacle plus every point within the radius	table 3.625 m^2 grows to 5.747 m^2
Minkowski sum	every point of one shape plus every point of the other	the board's region at 0°: 7 corners, at most 4 + 4
Slice	the obstacle region at one heading	the doorway is closed to the centre at 0°, open over 0.90 m at 90°
Connected component	a piece of free space that free paths can cross	disk: 1; board: 2 to 4 per slice, 1 in all

- For the disk robot a configuration is free when its centre lies outside every obstacle grown by 0.25 m: the centre (3.5, 3.2) is 0.2 m from the table, so it is in collision, the doorway leaves 0.7 m for the centre, and the straight 3.4 m drive to the doorway centre clears the grown table by 0.132 m, because the disk touches an obstacle exactly when its centre is within 0.25 m of it. The robot carrying the 1.4 m board must face the door, because the doorway is closed to its centre at 0° and open over 0.90 m at 90°.
- The **obstacle region** (G-2650) is drawn once in C-space, so a planner can test a configuration as a point instead of testing the whole robot against every obstacle.
- Growing a convex obstacle pushes its edges out and rounds its corners, so its area grows by perimeter times radius plus one circle, because the corner turns of a convex outline add up to 360°.
- The **Minkowski difference** (G-2654) $O \ominus A$, every obstacle point minus every robot offset, is the obstacle region of a robot that keeps its heading, because a robot point a lands on an obstacle point o exactly when the centre is at $o - a$; for robots symmetric about their centre, such as the disk and the board, it is the **Minkowski sum** (G-2653) of the obstacle and the robot.
- Two convex polygons give a convex sum with at most as many corners as both together, so the sum is quick to build from the corner sums and their convex hull.
- For a robot that is not round, each heading has its own slice of C-space, so the board must face the door: the doorway is shut to its centre at 0° and open over 0.90 m at 90°.
- A path exists only within one **connected component** (G-2656) of free space, so counting pieces tells before any search whether the door can be reached; turning joins the board's 2 to 4 pieces per slice into one.

The robot can now tell, for any configuration, whether its body is clear of the room, and the disk robot has a free straight drive to the door. Still missing: building the obstacle region exactly costs too much for real robots, so planners test one configuration or one segment at

a time, and they need those tests to be fast and their distances to treat headings correctly. Collision checking and nearest neighbours in C-space builds those tests.

6. Sources

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Other references

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- Shapely 2 documentation, `buffer` (growing a shape by a distance), https://shapely.readthedocs.io/en/s

7. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Obstacle region (C-obstacle) (G-2650)	The set of all configurations at which the robot's body touches or overlaps an obstacle, C_{obs}; drawing it in C-space lets a planner test a placement as one point, such as the disk robot's centre (3.5, 3.2), which lies in it 0.2 m from the table.
Free space (G-2651)	All configurations at which the robot touches no obstacle, C_{free}, the C-space without the obstacle region; a planner looks for a path that stays inside it.
Grown obstacle (inflated obstacle) (G-2652)	An obstacle with every point within the robot's radius added, its edges pushed out and its corners rounded; for a disk robot it is exactly the obstacle region, so the robot can be treated as a point (the table, 3.625 m^2, grows to 5.747 m^2 for a 0.25 m radius).
Minkowski sum (G-2653)	The set of all sums of a point of one shape and a point of the other, $A \oplus B$, like sweeping one shape over the other; the obstacle region of a robot that keeps its heading is the obstacle's Minkowski sum with the robot reflected through its centre.
Minkowski difference (G-2654)	The set of all differences of a point of one shape and a point of the other, $A \ominus B$, which equals $A \oplus (-B)$; obstacle minus robot gives the obstacle region, and two convex bodies overlap exactly when their difference contains the origin.
Connected component (of free space) (G-2656)	A piece of free space within which the robot can move from any configuration to any other without touching an obstacle; a path between two configurations exists only if both lie in the same component.
Configuration space (C-space) (G-2287)	The set of all configurations of a robot, one point per placement of the whole robot, such as a box of (x, y, θ) whose top and bottom faces are the same headings for a floor robot; motion planning looks for paths through it.

Term	Meaning
Configuration (G-2286)	The list of numbers that fixes the position of every point of a robot, such as $q = (x, y, \theta) = (2, 1, 0.524)$ for a floor robot or the joint angles (θ_1, θ_2) of a 2R arm; with it a planner treats the whole robot as one point.
Intersection (A ∩ B) (G-967)	The set of elements that are in both A and B, such as the points inside both of two half-planes; for events, the event that both A and B happen.
Euclidean distance (G-715)	The straight-line distance between two points.
atan2 (G-2325)	A function atan2(rise, run) that returns the direction of a line over the full circle from its rise and run given separately; needed because the plain inverse tangent of the ratio cannot tell opposite directions apart.
Convex set (G-479)	A set that contains the whole segment between any two of its points.
Convex hull (G-477)	The smallest region with straight edges that contains a set of points; every weighted average of the points lies inside it.